

15 Philippine institute of Pure and Applied Chemistry (PIPAC)

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Field of Testing :: Chemical Testing

Chemical Testing

Product	Specific Test	Method / Reference
I. Foods		
.01 Cereal and cereal products	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
	Total B-carotene	EN 12823-2 ICS 67.040 (2000) (PIPAC AP 018: High Performance Liquid Chromatography)

	Dietary Fiber	AOAC 992.16/985.29, 18th Ed. (PIPAC AP 041: Enzymatic-Gravimetric Method)
	Minerals: Calcium (Ca) Magnesium (Mg) Iron (Fe) Sodium (Na) Potassium (K) Zinc (Zn)	AOAC 984.27, 18th Ed. (PIPAC AP 055: Inductively Coupled Plasma Spectrometry)
.02 Nuts and nut products	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
.03 Dairy products	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
	Minerals: Calcium (Ca) Magnesium (Mg) Iron (Fe) Sodium (Na) Potassium (K) Zinc (Zn)	AOAC 984.27, 18th Ed. (PIPAC AP 055: Inductively Coupled Plasma Spectrometry)

	Omega-3 /Eicosapentaenoic acid (EPA) - Docosahexaenoic acid (DHA)	AOAC 991.39/922.06, 18th Ed. (PIPAC AP 029: Gas Chromatography)
.04 Meat and meat products	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
	Omega-3 /Eicosapentaenoic acid (EPA) - Docosahexaenoic acid (DHA)	AOAC 991.39/922.06, 18th Ed. (PIPAC AP 029: Gas Chromatography)
.05 Fish and fish products, including mollusks, crustaceans, and echinoderms	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066 Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)

	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
	Histamine	Journal of Food Science 41, 1282 (1976) PIPAC AP 002: Fluorometry)
	Mercury	AOAC 977.15, 18th Ed. (PIPAC AP 005: Cold Vapor Atomic Absorption Spectrophotometry)
	Heavy Metals: Cadmium (Cd) Chromium (Cr) Copper (Cu) Iron (Fe) Manganese (Mn) Nickel (Ni) Lead (Pb) Zinc (Zn) Magnesium (Mg)	AOAC 972.23, 18th Ed. AN 14082:2003 (PIPAC AP 006: Inductively Coupled Plasma Spectrometry)
	Omega-3 /Eicosapentaenoic acid (EPA) - Docosahexaenoic acid (DHA)	AOAC 991.39/922.06, 18th Ed. (PIPAC AP 029: Gas Chromatography)
.06 Sugar and sugar products	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)

.07 Confectioneries	Moisture	AOAC 925.45/934.06, 18th Ed.AOAC 934.01/920.155/925.10, 18th Ed.(PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066 Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
.08 Fruits, jams and other fruit products	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)

	Total B-carotene	EN 12823-2 ICS 67.040 (2000) (PIPAC AP 018: High Performance Liquid Chromatography)
.09 Vegetables and vegetable products	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total B-carotene	EN 12823-2 ICS 67.040 (2000) (PIPAC AP 018: High Performance Liquid Chromatography)
.10 Alcoholic beverages	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Moisture Content	USP <921> USP 37-NF 32 (2014) (PIPAC AP 022: Karl Fischer)
.11 Soft drinks and cordial	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)

	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
.12 Fruit juices, drinks and concentrates	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method)AOAC 922.06, 18th Ed.(PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
	Dietary Fiber	AOAC 992.16/985.29, 18th Ed. (PIPAC AP 041: Enzymatic-Gravimetric Method)
	Total B-carotene	EN 12823-2 ICS 67.040 (2000) (PIPAC AP 018: High Performance Liquid Chromatography)
	Glutathione	Official Monographs: Glutathione. USP 37-NF 32 (2014) (PIPAC AP 019: High Performance Liquid Chromatography)

.13 Edible fats and oils	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
	Moisture Content	USP <921> USP 37-NF 32 (2014) (PIPAC AP 022: Karl Fischer)
.14 Margarine	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
	Moisture Content	USP <921> USP 37-NF 32 (2014) (PIPAC AP 022: Karl Fischer)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed.AOAC 981.10/928.08, 18th Ed.(PIPAC AP 017: Modified Kjeldahl Method)

.15 Egg and egg products	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
	Omega-3 /Eicosapentaenoic acid (EPA) - Docosahexaenoic acid (DHA)	AOAC 991.39/922.06, 18th Ed. (PIPAC AP 029: Gas Chromatography)
.19 Other food additives	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
.20 Other food products	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)

	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
.24 Sauce, spices and condiments	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
	Total B-carotene	EN 12823-2 ICS 67.040 (2000) (PIPAC AP 018: High Performance Liquid Chromatography)
	3- Monochloropropane- 1,2-dioal esters (3- MCPD)	AOAC 2000.01, 18th Ed. CEN/TC 275 N 396 add (2000) (PIPAC AP 024: Gas Chromatography-Mass Spectrometric Detection)
.25 Food supplement and/or dietary supplement	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)

	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)
	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Dietary Fiber	AOAC 992.16/985.29, 18th Ed. (PIPAC AP 041: Enzymatic-Gravimetric Method)
	Minerals: Calcium (Ca) Magnesium (Mg) Iron (Fe) Sodium (Na) Potassium (K) Zinc (Zn)	AOAC 984.27, 18th Ed. (PIPAC AP 055: Inductively Coupled Plasma Spectrometry)
	Glutathione	Official Monographs: Glutathione. USP 37-NF 32 (2014) (PIPAC AP 019: High Performance Liquid Chromatography)
	Total B-carotene	EN 12823-2 ICS 67.040 (2000) (PIPAC AP 018: High Performance Liquid Chromatography)
	Omega-3 /Eicosapentaenoic acid (EPA) - Docosahexaenoic acid (DHA)	AOAC 991.39/922.06, 18th Ed. (PIPAC AP 029: Gas Chromatography)
.26 Novel food products	Moisture	AOAC 925.45/934.06, 18th Ed. AOAC 934.01/920.155/925.10, 18th Ed. (PIPAC AP 015: Gravimetry)
	Calories Carbohydrates	Food and Nutrition Research Institute, The Philippine Food Composition Table. 1997 AOAC, Method of Analysis for Nutritional Labeling, 1993 (PIPAC AP 066: Computation)

	Ash	AOAC 900.02/940.26, 18th Ed. AOAC 920.153/923.03, 18th Ed. (PIPAC AP 015: Gravimetry)
	Crude Protein	AOAC 935.58/981.10/928.08, 18th Ed. AOAC 981.10/928.08, 18th Ed. (PIPAC AP 017: Modified Kjeldahl Method)
	Total Fat	ISO 1143:1973-04-15 (Codex STAN Method) AOAC 922.06, 18th Ed. (PIPAC AP 016: Acid Hydrolysis Pet Ether Extraction, and Gravimetry)
II. Residual in Foods and Agricultural Materials		
.03 Antibiotics	Chloramphenicol	Reporter US Volume 32.1 Sigma Aldrich / EN 15662 (PIPAC AP 064: Liquid Chromatography-Tandem Mass Spectrometry)
III. Drugs and Pharmaceuticals		
.01 Drugs	Moisture Content	USP <921> USP 37-NF 32 (2014) (PIPAC AP 022: Karl Fischer)
.02 Herbal product		
.04 Medicinal and veterinary preparations		
.05 Vitamins		
.06 Antibiotics		
IV. Cosmetics, perfumes and essential oils		
.03 Essentials	Hydroquinone	Official Monographs: Hydroquinone.USP 37-NF 32 (2014)(PIPAC AP 020: High Performance Liquid Chromatography)
	Tretinoin	Official Monographs: Tretinoin . USP 37-NF 32 (2014) (PIPAC AP 021: High Performance Liquid Chromatography)
.04 Hygeine products	Identification and Determination of Salicylic Acid	In-house method PIPAC AP 010: (Visible Spectrometry)
	Identification and Determination of Sulfur Content	In-house method PIPAC AP 009: (Gravimetry)

	pH	4500-H ⁺ B, SMEWW, 22 nd Ed. / PNS 23:2002 (PIPAC AP 007: pH Meter)
	Aluminum	AOAC 938.11, 18th Ed. (PIPAC AP 023: Inductively Coupled Plasma Spectrometry)
V. Detergents and related products		
.01 Soap	pH	4500-H ⁺ B, SMEWW, 22 nd Ed. / PNS 23:2002 (PIPAC AP 007: pH Meter)
.02 Synthetic detergents		
.05 Disinfectants		
VI. Water		
.01 Bottled water	pH	4500-H ⁺ B, SMEWW, 22 nd Ed. / PNS 23:2002 (PIPAC AP 007: pH Meter)
	Arsenic (As)	3114B, SMEWW, 22 nd Ed. (PIPAC AP 011: Atomic Absorption Spectrophotometry)
	Conductivity	2510B, SMEWW, 22 nd Ed. (PIPAC AP 008: Conductivity Meter)
	Total Dissolved Solids (TDS)	2540C, SMEWW, 22 nd Ed. (PIPAC AP 013: Gravimetry)
	Mercury	SMEWW 3112B (PIPAC AP 057: Cold-Vapor Atomic Absorption Spectrometry)
	Hexavalent Chromium (Cr ⁶⁺) Heavy Metals: Calcium (Ca) Cadmium (Cd) Chromium (Cr) Copper (Cu) Iron (Fe) Potassium (K) Manganese (Mn) Sodium (Na) Nickel (Ni) Lead (Pb) Zinc (Zn) Magnesium (Mg)	350-Cr B, SMEWW, 22 nd Ed. (PIPAC AP 014: UV-Vis) 3030A, E/3120B, SMEWW, 22 nd Ed. (PIPAC AP 026: Inductively Coupled Plasma)

	Benzene	US-EPA 5021/8021B (PIPAC AP 065: Head Space Chromatography)
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Legend to Reference Methods:

AOAC - Association of Official Analytical Chemists

ISO - International Organization for Standardization

SMEWW - Standard Methods for the Examination of Water and Wastewater

US-EPA - United States Environmental Protection Agency

USP - United States Pharmacopeia

****Nothing Follows****